

Effective AI Prompting: Your Essential Skill

Madhulika Goyal
IIT Bombay

04 December 2025



Spoken Tutorial
Developed at IIT Bombay

Effective AI Prompting: Your Essential Skill

- Welcome to this Spoken Tutorial on Effective Prompting for AI.



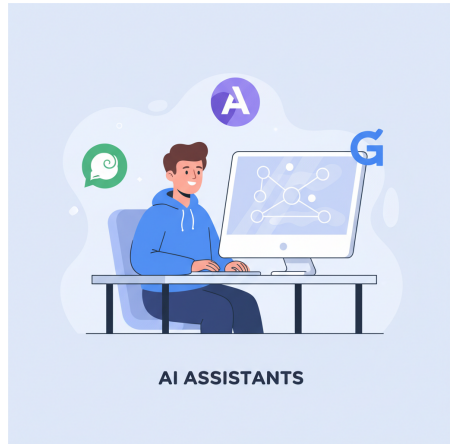
Learning Objectives

- Recall the three core components of an effective AI prompt.
- Apply advanced techniques like Chain-of-Thought and structured output.



Prerequisites

- Basic computer literacy.
- Access to a Large Language Model (LLM) tool.
- Examples: Google Gemini, ChatGPT, or Claude.



AI as a Smart Intern: The Power of Prompting

- AI models are like smart, fast interns.
- They need clear instructions to do good work.
- Prompting is giving these instructions.
- A good prompt



The 3 C's: Clarity

- Good prompts almost always include: Clarity, Context, and Constraints.
- Clarity: Be specific!
AI does not know what you are thinking.



The 3 C's: Context and Constraints + Persona

- Context: Give background. Tell AI why and for whom.
- Better Prompt (with Context):
"Write for a sustainable lifestyle"



Spoken Tutorial
Developed at IIT Bombay

Giving Examples: Zero, One, and Few-Shot

- Zero-Shot: You give zero examples. (Most common).
- One-Shot: You give one example to show a pattern.
- Example:
"Summarize text"



Advanced Techniques: Chain-of-Thought (CoT)

- Chain-of-Thought (CoT): Ask AI to "Think Step-by-Step".
- Useful for logic, math, reasoning problems.
- Reduces errors by



Advanced Techniques: Structuring Output & Iterative Prompting

- Structuring Output: Tell AI the exact format you need (Markdown table, HTML, JSON, list).
- Prompt: "List pros and cons of remote



Practical Lab: Text Generation Challenge

- Open Google Gemini, ChatGPT, or Claude.
- Try this prompt: "Act as a fitness coach. Create a simple 3-day workout plan for a beginner who wants to exercise at home with no equipment. The goal is a full-body workout. Present it in a table format with columns for 'Day', 'Exercise', and 'Duration'."



Spoken Tutorial
Developed at IIT Bombay

Practical Lab: Image Prompting Challenge

- Image tools like Midjourney or DALL-E 3 use descriptive prompts.
- Think like an artist: describe visual details.
- Try this: "A close-up, photorealistic image of a cat wearing a tiny detective hat and magnifying glass, sitting in a library. The lighting is warm and cinematic."
- Use Midjourney or DALL-E 3 if available.



Spoken Tutorial
Developed at IIT Bombay

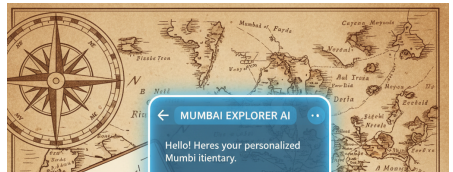
Summary

- AI needs specific instructions (prompts).
- Good prompts have Clarity, Context, Constraints (3 C's).
- Use Persona and examples (Zero, One, Few-Shot).



Assignment

- Use an AI tool (Gemini/ChatGPT/Claude).
- Prompt: "Act as a friendly tour guide for Mumbai. Create a 1-day itinerary for a family visiting Mumbai for the



Thank You!

- Spoken Tutorials are a project by EduPyramids / FOSSEE, IIT Bombay.
- More information at spoken-tutorial.org.
- Thank you for

